Drug and Alcohol REVIEW

APSAD

Drug and Alcohol Review (May 2016), 35, 359–366
DOI: 10.1111/dar.12302

The hidden costs of drug and alcohol use in hospital emergency departments

KERRYN BUTLER[1], REBECCA REEVE[2,3], SHEENA ARORA[2], ROSALIE VINEY[2], STEPHEN GOODALL[2], KEES VAN GOOL[2] & LUCINDA BURNS[1]

[1]*National Drug and Alcohol Research Centre, UNSW Australia, Sydney, Australia,* [2]*Centre for Health Economics Research and Evaluation, University of Technology, Sydney, Australia, and* [3]*Centre for Social Impact, UNSW Australia, Sydney, Australia*

Abstract

Introduction and Aims. *This study estimates the burden of drug and alcohol morbidity on hospitals in New South Wales (NSW) by observing a multi-site collective sample utilising survey information and data linkage. Specifically we aimed to determine the prevalence of alcohol and other drug (AOD) problems and to estimate patterns of utilisation of hospital services, costs of presentations, and admissions for patients with AOD problems.* ***Design and Methods.*** *Patients were recruited from eight NSW public hospitals presenting to the hospital emergency department over a 10 day period. Participants completed a self-administered survey with demographic characteristics and questions about substance use. More than two-thirds (68%) of participants consented to provide access to their NSW Health medical data for a period spanning 2.5 years.* ***Results.*** *One-third (35%) of the total sample were identified as having problematic AOD use with one in five of these patients requiring a high level of intervention. Those patients requiring a high level of intervention present more often and cost more per presentation. If admitted they were more likely to have longer stays and were also more likely to be admitted to a psychiatric ward and have a longer stay in the ward.* ***Discussion.*** *This study demonstrates a need for AOD interventions in the emergency department setting, both because it represents an opportunity for intervention in a population in which problems with substance use is highly prevalent, and because there is evidence that AOD imposes additional costs on the health system.* [Butler K, Reeve R, Arora S, Viney R, Goodall S, van Gool K, Burns L. The hidden costs of drug and alcohol use in hospital emergency departments. *Drug Alcohol Rev* 2016;35:359–366]

Key words: alcohol, drug, emergency department, comorbidity, hospital presentation.

Introduction

Comorbidity between alcohol and other drug use is now common leading to a high prevalence of alcohol and other drugs (AOD) morbidity among patients presenting in emergency departments (ED). AOD morbidity is known to affect postoperative morbidity [1], behavioural incidents, re-admission and re-injury rates [2]. Despite the high prevalence, AOD morbidity is frequently unidentified at the time of presentation or on admission [3,4]. Timely identification of patients presenting to hospital who have an AOD problem, regardless of presenting complaint, is necessary to determine possible contraindications in treatment and to inform treatment pathways. Understanding the prevalence and cost of patients presenting to hospital is key to defining the size of AOD issues and to establish the case for intervention in this patient group.

Previous studies by Conigrave *et al*. [5] and Egerton-Warburton *et al*. [6] examined alcohol presentations and found prevalence rates ranging from 5% to 41% for problem drinkers presenting to Australian EDs. A further study by Indig *et al*. [7] also included illicit drugs; however, this was a retrospective study using nursing triage text to detect AOD ED presentations in New South Wales (NSW), Australia.

Kerryn Butler MSocHlth, Research Officer, Rebecca Reeve B.Ec(Hons), PhD, Senior Research Fellow, Sheena Arora BSc, BCom, MPH, Research Fellow, Rosalie Viney B.Ec, M.Ec, PhD, Professor of Economics, Stephen Goodall BSc, MSc, PhD, Associate Professor, Kees van Gool M.Ec, PhD, Associate Professor, Lucinda Burns MPH, PhD, Associate Professor. Correspondence to Ms Kerryn Butler, National Drug and Alcohol Research Centre, UNSW, Sydney, NSW 2052, Australia. Tel: +61 2 9385 0187; Fax: +61 2 9385 0222; E-mail: kerryn.butler@unsw.edu.au

Received 1 December 2014; accepted for publication 20 May 2015.

This study is the first in Australia that estimates the burden of AOD morbidity on hospitals in NSW by observing a multi-site collective sample. Specifically we aimed to:

- Determine the prevalence of AOD problems among patients presenting to EDs.
- Estimate patterns of utilisation of hospital services and costs of presentations and admissions for patients with AOD problems, compared with other patients over a 2.5 year period.

Methodology

Study design

Patients were recruited from eight NSW public hospitals presenting to the hospital ED 24 h a day over 10 consecutive days. Consenting participants completed a self-administered survey with demographic characteristics and questions about substance use. A subsample consented to provide access to their NSW Health admitted patient records. The survey was cross-sectional in nature, with a single 10 day period in each hospital. Ethics approval was obtained from NSW Population and Health Services Research Ethics Committee, and site-specific assessments were obtained for each individual hospital where data were collected.

Procedure

All potentially eligible patients presenting to each ED were approached and invited to participate. Eligibility criteria were: 16 years of age or older, showed no sign of trauma requiring urgent medical attention, showed no signs of openly aggressive behaviour and appeared to be able to make an informed decision (as assessed by the research interviewer). Individuals who were severely intoxicated, cognitively impaired or with insufficient English language skills to complete the questionnaire were excluded. Consequently, estimates from this study are likely to be conservative, representing a lower bound of the size of burden of AOD problems in NSW EDs. Research staff trained in issues of confidentiality, interviewing and consenting skills were stationed in the ED of each hospital. Informed consent by way of a consent form was obtained before handing out a self-administered questionnaire, which was collected 20–30 min later. In instances where the participants were unable to complete the questionnaire themselves (patients with injuries preventing them from writing), a research team member provided assistance.

Instrument

The questionnaire measured: reason for presentation, self-perceived contribution of substance use to current presentation, substance use in past 24 h, recent problematic substance use [as assessed by the Alcohol, Smoking and Substance Involvement Screening Test (ASSIST)] [8], general functioning [as assessed by the World Health Organization Disability Assessment Schedule (WHODAS) 2.0] [9] and use of AOD services.

The World Health Organization's ASSIST is a tool designed to detect risky or hazardous use, including but not limited to dependence. The tool is designed to be used in primary and general medical care settings. The ASSIST screens for problem or risky use of tobacco, alcohol, cannabis, cocaine, amphetamine-type substances, sedatives, hallucinogens, inhalants, opioids and 'other' drugs. A simple scoring method is used to determine the most appropriate intervention ('no treatment', 'brief intervention' or 'referral to specialist assessment and/or more intensive treatment').

The WHODAS is an assessment instrument used to measure general functioning and level of disability. A simple scoring method is used to calculate a value, used to describe the degree of disability, where 12 represents no disability and 60 represents full disability.

Data linkage

All survey participants were invited to provide separate consent to NSW Health data linkage and more than two-thirds (68%) did so. Survey data for this subsample were linked to NSW Admitted Patient (AP) records and ED records by the Centre for Health Record Linkage. ED and AP records were obtained for the period from 18 months prior to the survey up to 1 year post survey.

Analysis

Statistical analysis of survey data was performed using SPSS for Windows, version 22 (IBM Corp, Armonk, NY, USA) [10]. Descriptive statistics were used to compare AOD presentations with non-AOD presentations. The AOD group in this study was those who: (i) scored positive for substance use on the ASSIST; and (ii) were then classified as being in need of some level of intervention for that substance. Those who scored positive for tobacco *only* were not included in the AOD group for analysis. Differences between means were examined using *t*-test, and between proportions using Pearson's χ^2 test. When comparing proportion of males in the AOD and the non-AOD groups, a Levine's test indicated unequal variances ($F = 48.228$, $P < 0.001$), so degrees of freedom were adjusted from 1568 to 1331.

Analysis of the survey data linked to 2.5 years of ED and AP data was conducted in STATA version 12.1 (StataCorp LP, College Station, TX, USA) [11]. Descriptive comparisons of ED and AP utilisation and costs were undertaken between patients in the AOD and non-AOD group, as defined above. The AOD group was divided according to level of intervention need (brief or specialist treatment). We compared: frequency of presentations and admissions; average cost of presentations and admissions; time and day of week of arrival; average length of stay; time in a psychiatric ward; and time in an intensive care unit (ICU). Frequency and costs were analysed per quarter rather than per annum as the data do not span a whole number of years. Costs of presentations and admissions were derived from the NSW Costs of Care Standards 2010–2011 [12] and inflated to 2012 prices using the Australian Institute of Health and Welfare price index for 'government final consumption expenditure on hospitals and nursing homes' [13]. Overall differences between means were examined using analysis of variance *F*-tests, and between proportions using Pearson's χ^2 test. Pairwise differences were examined using *t*-tests; all cell sizes were large (>100), and therefore *t*-tests are valid according to the 'central limit theorem'. As there were multiple observations per individual in the ED and AP data, the standard errors were adjusted for clustering.

Results

Of 4132 ED patients (including infants and children who were out of scope for this study), 3658 potentially eligible individuals were approached to complete the survey. Of this 3658, 17% ($n = 615$) were ineligible and 32% ($n = 1184$) declined to participate. Patients were approached in the ED waiting room after they had been triaged. Acknowledging that patients who are feeling ill may not want to voluntarily participate in a survey, reasons for declining to participate were not sought. Thus, a response rate of 51% ($n = 1859$) was achieved. Due to 244 patients being transferred for treatment prior to the collection or completion of the surveys, the final total sample comprised of 1615 completed and returned surveys. The initial analysis was done on the sample of 1615 with the exception of the AOD grouping (AOD and Non-AOD), which was done on $n = 1574$ due to insufficient ASSIST data to allocate those cases to an AOD group. Further analysis was done on a subsample of 1078 participants who consented to NSW ED and AP data linkage and provided sufficient ASSIST data to be allocated an AOD group status.

Demographics of total sample

Overall, the sample was comprised of 53% male and 47% female. The age of participants ranged from 16 to 98 years of age ($M = 41.16$, $SD = 18.96$). Five percent of participants identified as being of Aboriginal or Torres Strait Islander origin. Male participants were not significantly older than female participants (Males; $M = 41.52$, SD 18.9 years vs. Females; $M = 40.75$, SD 18.9 years, $P > 0.5$).

Substance use among total sample

Among the total sample 35% of participants self-reported substance use in the past 24 h and 30% reported their substance use had contributed to their current hospital presentation. Alcohol was the most common substance reported being used in the 24 h prior to presentation with 27% of the sample reporting this use. Other substances commonly reported being used in the 24 h prior to presentation were sedatives (5%), cannabis (4%), opioids (4%) and amphetamine-type stimulants (2%). Participants who reported that their substance use had contributed to their current presentation most commonly reported alcohol (18%), cannabis (4%), sedatives (3%), amphetamine-type stimulants (2%) and opioids (2%) as contributing to their presentation.

One-third (35%) of the total sample screened positive for risky substance use in need of some level of intervention (either 'brief intervention' or 'specialist treatment'), as measured by the ASSIST. This group is forthwith referred to as the AOD group.

Ninety-one per cent of the AOD group produced a Specific Substance Involvement Risk Score, for one or more substances, which indicated the need for a 'brief intervention'. Twenty-one per cent of the AOD group produced a Specific Substance Involvement Risk Score, for one or more substances, which indicated the need for a specialist treatment. Polydrug substance use requiring intervention was common with 35% of cases in the AOD group requiring intervention for two or more substances concurrently. Substances that commonly required intervention included alcohol, cannabis, illicit use of sedatives and amphetamine-type stimulants (see Figure 1).

Comparing the AOD and non-AOD groups

Compared with the non-AOD group, the AOD group were more likely to be male ($P < 0.001$), younger in age and have a higher disability score.

In the AOD sample, 63% were male compared with 49% of the non-AOD group being male. A χ^2 test revealed that this difference was statistically significant, χ^2 (1) = 26.38, $P = < 0.001$. An independent-samples *t*-test indicated that the AOD group was significantly younger ($M = 36.89$, $SD = 16.328$), $t(1331) = 6.87$, $P < 0.001$, $d = 0.34$. Although a higher proportion of people in the AOD group identified as having

Aboriginal or Torres Strait Islander origins compared with the non-AOD group, this difference was not found to be statistically significant.

The WHODAS group median scores showed that the AOD group had statistically higher levels of disability compared with the non-AOD group. A Mann–Whitney test indicated that there is a statistically significant difference between the distributions of the disability scores of the AOD group ($Mdn = 16$) and the non-AOD group ($Mdn = 13$), $U = 192392$, $P < 0.001$, $r = 0.36$. This statistical difference was still observed when age was grouped into categories (<29-years-old, 30- to 48-years-old and >49-years-old).

Subsample with data linkage

Of the subsample with data linkage, 36% screened positive for risky substance use in need of some level of intervention (either 'brief intervention' or 'specialist treatment'), as measured by the ASSIST, compared with 35% of the overall sample. This subsample is used for comparison of resource use and costs by AOD status and level of intervention need, according to the ASSIST score.

ED presentations

There were 5692 presentations over 2.5 years for the 1078 people in the subsample. This equates to an average of 0.53 presentations per person per quarter over the observation period. Men presented less frequently than women (46% of presentations were for men who comprise 53% of the subsample) and Australian-born people presented slightly more frequently than other cultural groups (84% of presentations were for Australian born, who comprise 79% of the subsample). The mean age per presentation is consistent with the mean age of the subsample, 41. The overall mean number and cost of presentations per person per quarter, by ASSIST screen status, is presented in Table 1.

Overall, the number of presentations per person per quarter, over the 2.5 year observation period, differed significantly by ASSIST screen status. People identified

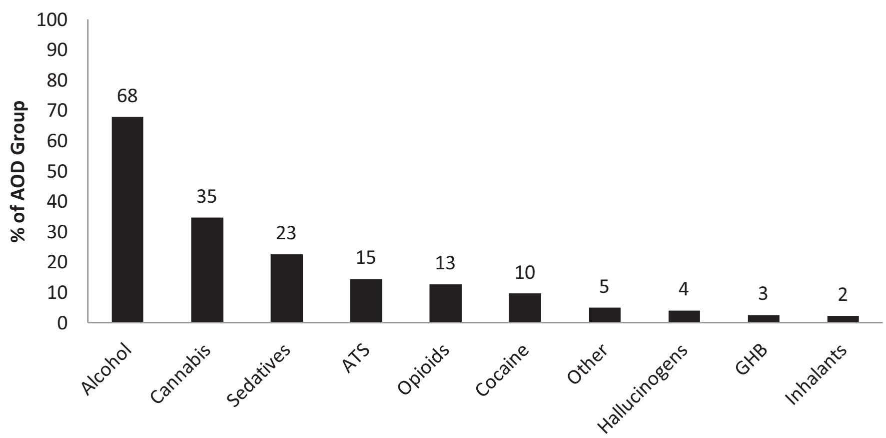

Figure 1. *Proportion of AOD group who require intervention for substance use, by substance type.*

Table 1. *Average cost of presentations per patient per quarter by ASSIST screen status*

ASSIST screen status	Mean number of presentations	Mean cost of presentations ($)
Overall ($n = 10\ 780$)	0.53 (0.45–0.61)	210.56 (180.17–240.95)
ASSIST negative ($n = 6790$)	0.47 (0.36–0.58)	188.00 (146.61–229.39)
ASSIST positive: brief intervention ($n = 3020$)	0.53 (0.43–0.62)	211.73 (172.21–251.24)
ASSIST positive: specialist intervention ($n = 970$)	0.94 (0.63–1.25)	364.83 (246.83–482.84)
Test for overall significant difference by ASSIST screen status	$F(2,1076) = 3.95$, $P = 0.020$	$F(2,1076) = 3.86$, $P = 0.021$

Notes: $n = 10\ 780$ represents 10 quarters of observations for 1078 individuals. Sample includes 13 individuals with zero presentations. 95% confidence intervals in parentheses. ASSIST, alcohol, smoking and substance involvement screening test.

in the survey as requiring specialist intervention presented twice as often as those who screened negative for AOD problems (*t*-test $P = 0.005$). The mean costs of presentations per person per quarter also differed significantly by ASSIST screen status, with people requiring a specialist intervention at the time of the survey, costing an additional \$177 per quarter than those without AOD problems (*t*-test $P = 0.006$). Pairwise differences between those needing brief intervention and the non-AOD group were not significant. We also examined the length of stay in ED (from triage to departure) but found no significant difference by ASSIST screen status.

Time of presentations by ASSIST screen status

Over the 2.5 year observation period, patients presented to ED at all hours of the day and night, with the fewest presentations in the hours after midnight and more presentations occurring in the afternoon/evening than at other times of day. Peak presentation time for patients who screened negative for AOD problems is between 6 pm and 9 pm. Patients who screened in the survey as requiring specialist intervention were more likely to present either in the afternoon or slightly later in the evening. Differences by ASSIST screen status were not significant at the 5% level ($P = 0.08$).

Inpatient admissions

This analysis included all AP episodes for consenting participants, including NSW Public Hospitals, Public Psychiatric Hospitals, Public Multi-Purpose Services, Private Hospitals and Private Day Procedures Centres. There were a total of 3483 admissions over 2.5 years. This equates to an average of 0.32 admissions per person per quarter over the observation period. Men were admitted less frequently than women (48% of admissions were for men who comprise 53% of the subsample), and Australian-born people were admitted slightly more frequently than other cultural groups (83% of admissions were for Australian born, who comprise 79% of the subsample). The mean age per admission was 50, which is older than the mean age of the subsample of 41. The overall mean number and cost of admissions per person per quarter, by ASSIST screen status, is presented in Table 2.

The mean number of admissions per person per quarter for the overall sample was 0.32, and the mean cost of admissions per person per quarter was \$1538.90. The number of admissions and the mean cost of admissions did not differ significantly by ASSIST screen status.

Utilisation of psychiatric wards and ICUs

Table 3 summarises the proportion of admissions that spent time in a psychiatric ward or ICU, by ASSIST screen.

Overall, approximately 11% of admissions spent time in a psychiatric ward. Compared with those who screened negative, a significantly higher proportion of admissions where patients screened as requiring specialist intervention (*t*-test $P = 0.003$) spent time in a psychiatric ward. Of all admissions, only 2.15% of admissions spent time in an ICU. This did not vary significantly by ASSIST screen status.

Average length of stay

Table 4 summarises the average length of stay overall, in a psychiatric ward and in ICU for the 720 individuals with any admissions during the 2.5 year analysis period.

Table 2. *Average cost of admissions per quarter per person by ASSIST screen status*

ASSIST screen status	Mean number of admissions	Mean cost of admissions (\$)
Overall (*n* = 10 780)	0.32 (0.26–0.38)	1538.90 (1243.38–1834.41)
ASSIST negative (*n* = 6790)	0.32 (0.23–0.40)	1392.50 (1140.28–1644.72)
ASSIST positive: brief intervention (*n* = 3020)	0.32 (0.23–0.41)	1810.65 (944.70–2676.60)
ASSIST positive: specialist intervention (*n* = 970)	0.39 (0.26–0.51)	1717.56 (1098.09–2337.04)
Test for overall significant difference by ASSIST screen status	F(2,1076) = 0.48 P = 0.617	F(2,1076) = 0.78 P = 0.456

Notes: *n* = 10 780 represents 10 quarters of observations for 1078 individuals. Sample includes 358 individuals with zero admissions. 95% confidence intervals in parentheses. ASSIST, alcohol, smoking and substance involvement screening test.

Table 3. *Proportion of admissions that spent time in a psychiatric ward or intensive care unit, by ASSIST screen status*

Characteristic	All admissions $n = 3483$	Screened negative $n = 2141$	Screened positive: brief intervention $n = 966$	Screened positive: specialist intervention $n = 376$	Test for significant difference across categories
Admitted to psychiatric ward (%)	10.82 (7.59–15.21)	7.38 (3.86–13.64)	12.63 (6.78–22.31)	25.80 (16.06–38.71)	$F(1.85, 1330.93) = 4.9330$ $P = 0.009$
Admitted to intensive care unit (%)	2.15 (1.52–3.04)	2.24 (1.38–3.63)	2.17 (1.30–3.63)	1.60 (0.73–3.45)	$F(1.82, 1309.71) = 0.2110$ $P = 0.789$

Note: $n = 3483$ presentations for 720 individuals. 95% confidence intervals in parentheses. ASSIST, alcohol, smoking and substance involvement screening test.

Table 4. *Admissions characteristics by ASSIST screen status*

Characteristic:	All admissions $n = 3483$	Screened negative $n = 2141$	Screened positive: brief intervention $n = 966$	Screened positive: specialist intervention $n = 376$	Test for significant difference across categories
Length of stay (days)	4.08 (3.49–4.66)	3.57 (2.78–4.36)	4.54 (3.80–5.28)	5.78 (4.42–7.15)	$F(2, 718) = 4.13$ $P = 0.017$
Length of stay in a psychiatric ward (days)	1.02 (0.56–1.48)	0.64 (0.03–1.25)	1.22 (0.51–1.94)	2.66 (1.31–4.01)	$F(2, 718) = 3.70$ $P = 0.025$
Mean duration in ICU (hours)	1.19 (0.68–1.69)	1.05 (0.41–1.69)	1.08 (0.53–1.67)	2.17 (-0.38–4.73)	$F(2, 718) = 0.35$ $P = 0.705$

Note: $n = 3483$ presentations for 720 individuals. 95% confidence intervals in parentheses. ASSIST, alcohol, smoking and substance involvement screening test; ICU, intensive care unit.

The mean length of stay for all admissions was 4.08 days. Admissions for those with a positive screen status, requiring specialist intervention had a significantly longer mean length of stay compared with those with a negative screen status (*t*-test $P = 0.01$). The mean days spent in a psychiatric ward includes admitted patients who did not spend any time in a psychiatric ward. Therefore, the difference in length of stay can be partly explained by the higher proportion of admissions to a psychiatric ward in this group and partly explained by a longer length of stay among those admitted to a psychiatric ward (8.6 days in the screen negative group and 10.3 days in the specialist intervention group). The average time spent in ICU for all admissions was 1.19 h. This did not vary significantly by screen status.

Limitations

This study has a number of limitations. Patients who were too intoxicated to provide consent were excluded; this implies that our results may underestimate the burden of AOD on NSW EDs. However, our prevalence estimates are within the range reported by other studies. We also relied on self-report, and data may therefore be limited by recall and social desirability bias. Due to the voluntary nature of the study, bias may exist if there are systematic differences between people who respond and people who do not. The relatively low (51%) patient participation rate is an issue most likely to be faced by surveys conducted among hospital patients (even more so by attendees to EDs) as most patients are acutely sick or undergoing tests and disinclined to participate in or complete surveys. The analysis is descriptive and does not control for patient characteristics. However, men presented and were admitted less frequently than women, and younger people were less likely to be admitted; this suggests that our findings are conservative, given that AOD group are younger and more likely to be male than the non-AOD group.

Discussion

This study is the first Australian study to provide longitudinal data on health service utilisation of Australian

ED attenders with problematic alcohol and/or drug use by using data linkage to NSW Health record data.

More than one-third of patients presenting to ED screened positive for problematic substance use requiring some level of intervention, regardless of the reason for presentation. Even though AOD may not be the primary reason for presentation, the fact that those who screened as requiring specialist intervention present twice as frequently as patients with no AOD problems, over our 2.5 year observation period, indicates that AOD is a driver of hospital utilisation and higher cost to EDs. When admitted, people who screened as needing specialist intervention have a longer total length of stay compared with those who screened negative. These patients are also more likely to be admitted to a psychiatric ward during their stay and have a longer length of stay in the ward. This indicates a need for AOD interventions in an ED setting, both because it represents an opportunity for intervention in a population in which problems with substance use is highly prevalent, and because there is evidence that AOD imposes additional costs on the health system. As well as helping such patients to receive appropriate treatment, for those in need of specialist intervention, successful interventions may prevent repeat presentations and associated costs [14]. Like other patients, those with AOD problems present in the afternoon and evening suggesting that these would be appropriate times to target hospital-based intervention services.

Services targeted at addressing these problems, such as Alcohol and Other Drug consultation liaison services, are available in most NSW hospitals on an ad hoc basis; however, in many instances, these services are limited. AOD consultation liaison services provide specialist AOD services within hospital settings, specifically consultation (advice regarding the management of AOD-related issues for referred patients) and liaison (enhancing capacity of generalist health providers to address AOD issues in their routine clinical work). A recent report published online [15] revealed consultation liaison-type services can prevent an increase in average length of stay in ED over time, prevent a worsening in emergency admission performance, reduce the frequency of ED presentations over time, decrease the rate of admissions over time and provide net benefit and savings when implemented.

Drug and alcohol morbidity among patients presenting in EDs poses complex challenges for the patients, the staff and the hospitals. This paper has presented descriptive evidence of the potential burden of AOD on NSW hospitals. The findings presented in this paper are likely to be conservative but identify potential opportunities for providing interventions that can alleviate burdens and costs. Future work will build on this foundational work to investigate differences and changes over time between patient groups, including those who receive AOD consultation liaison services, controlling for confounding factors.

Conflict of interest

This paper presents findings from an evaluation of Consultation Liaison Services within NSW Hospitals. This work was funded by NSW Health. The current paper design, conduct and interpretation of findings are the work of the investigators; the funders had no role in this paper.

Acknowledgements

This study was funded by the New South Wales Department of Health.

References

[1] Tønnesen H. Alcohol abuse and postoperative morbidity. Dan Med Bull 2003;50:139–60.
[2] Rostenberg P. Alcohol and other drug screening of hospitalized trauma patients. Rockville, MD: Substance Abuse and Mental Health Services Administration, Center for Substance Abuse Treatment, Treatment Improvement Protocol #16, 1995. Available at: https://roar.nevadaprc.org/system/documents/994/original/TIP16.pdf?1349367795 (accessed September 2014).
[3] Shourie S, Conigrave KM, Proude EM, Ward JE, Wutzke SE, Haber PS. Pre-operative screening for excessive alcohol consumption among patients scheduled for elective surgery. Drug Alcohol Rev 2007;26:119–25.
[4] Indig D, Copeland J, Conigrave KM, Rotenko I. Attitudes and beliefs of emergency department staff regarding alcohol-related presentations. Int Emerg Nurs 2009;17:23–30.
[5] Conigrave KM, Burns FH, Reznik RB, Saunders JB. Problem drinking in emergency department patients: the scope for early intervention. Med J Aust 1991;154:801–5.
[6] Egerton-Warburton D, Gosbell A, Wadsworth A, Fatovich D, Richardson D. Survey of alcohol-related presentations to Australasian emergency departments. Med J Aust 2014;201:584–7.
[7] Indig D, Copeland J, Conigrave KM, Arcuri A. Characteristics and comorbidity of drug and alcohol-related emergency department presentations detected by nursing triage text. Addiction 2010;105:897–906.
[8] World Health Organization (WHO) ASSIST Working Group. The Alcohol, Smoking and Substance Involvement Screening Test (ASSIST) WHO. INT, 1992. Available at: http://www.who.int/substance_abuse/publications (accessed October 2013).
[9] Üstün TB. Measuring health and disability: manual for WHO disability assessment schedule WHODAS 2.0. Geneva: World Health Organization, 2010.
[10] IBM. SPSS for windows version 22.0 (Version 22.0). New York: IBM Corp., 2013.
[11] StataCorp. Stata statistical software: release 12. College Station, TX: StataCorp LP, 2011.
[12] NSW Health. NSW costs of care standards 2009–2010. Canberra: Inter-Government & Funding Strategies Branch, NSW Health Department, 2011:15.

[13] Australian Institute of Health and Welfare. Health expenditure Australia 2011–12. Health and welfare expenditure series no. 50. Cat. No. HWE 59. Canberra: AIHW, 2013.
[14] Crawford MJ, Patton R, Touquet R, *et al*. Screening and referral for brief intervention of alcohol-misusing patients in an emergency department: a pragmatic randomised controlled trial. Lancet 2004;364:1334–9.
[15] NSW Health. Evaluation of NSW Drug and Alcohol Consultation Liaison Services—Final Report, 2014. Available at: http://www.health.nsw.gov.au/mhdao/programs/da/Documents/rpt-eval-nswda-cl-serv.pdf (accessed 14 May 2015).